

Assessing agroforestry options for sustainable land use in Germany

EURAF 2022
Agroforestry for the Green Deal transition.
Research and innovation towards the
sustainable development of agriculture and
forestry

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Theme: Agroforestry business environment models

Keywords: Agroforestry, Decision Analysis, decision-making, modelling, probabilistic simulation, Germany

Abstract

Introducing trees to farming landscapes represents a risky decision with large uncertainties. In this project we worked with Woodify, a startup that reforests damaged woodlands and sells the CO₂ certificates. Woodify offers local carbon offsets through contracts with German forest owners. They bought a plot of land for reforestation in the Eifel region of Germany but could not use it for forest, as the land use plan did not allow it. However, the land use plan would allow for an agroforestry system where agricultural production is combined with trees. We applied holistic modeling processes to support Woodify in determining if such an investment would be economically viable.

Decision Analysis (DA) can be effective in informing such complex decisions. DA includes a set of methods that are applied for supporting well-informed decision-making. It involves identifying and assessing all the important aspects of a decision. The ultimate aim is to inform decision-making so that any actions based on the decision will be most likely to produce a preferred favorable outcome. DA aims to use all relevant and available information to support decision-making processes. It gives decision-makers the option to evaluate and model potential decision outcomes and thereby determine the best course of action based on expected outcomes. DA methods facilitate the development of a holistic understanding of the goals and intended impacts of the decision and the uncertainties involved. The process starts by framing the decision problem of interest and formulating it as an impact pathway. This pathway is then translated to a formal model, most input variables for which are expressed by probability distributions for their true values. This model is run many times in a probabilistic simulation, with variable inputs that are randomly drawn from the specified distributions. The resulting outcome distributions can then be compared across different decision options (e.g. comparing outcomes for agroforestry and treeless agriculture).

We worked with Woodify to model five possible combinations of hazelnut orchards, other trees, truffles, hay and chickens compared to a baseline of arable land with crop rotations of winter barley, rye, wheat, and summer barley. The outcomes of interest included farm profits, environmental benefits, and sequestered CO₂ (Figure 1). We programmed this as a probabilistic model to calculate the costs, benefits and risks of the decision and the effects on the expected outcomes over 30 years. Income from hazelnuts, eggs and truffles were considered to be the main sources of farm benefits. Major costs included establishment costs, e.g. setting up fences, and maintenance costs, e.g. labor costs. Risks included the need for additional labor and inputs such as those used to deal with outbreaks of truffle infections.

We subtracted the results of the baseline model from modeled outcomes for the agroforestry scenarios. All model results showed a positive outcome after 30 years compared to maintaining traditional crop rotations. This indicates that the decision to convert the land to agroforestry production is a good one, which is likely to pay off in the long term.